Fiscal Year:	FY 2012	**Task Last Updated:**	FY 06/27/2012
PI Name:	Wheeler, Richard M.S.		
Project Title:	Multiphase Transport in Porous Media: Gas-Liquid Phase Separation Using Capillary Pressure Gradients – ISS Experiment Development		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	97457-0102	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	07/01/2009	**End Date:**	06/30/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: PI change in mid-2009 from James Atwater to Richard Ray Wheeler (per PI and NSSC 4/2010)		
Key Personnel Changes/Previous PI:	NOTE: PI change in mid-2009 from James Atwater to Richard Ray Wheeler (4/2010)		
COI Name (Institution):	Jovanovic, Goran (J Lab & Associates)		
Grant/Contract No.:	NNX09AK64G		
Performance Goal No.:			
Performance Goal Text:			
	We propose the development of a series of International Space Station (ISS) microgravity experiments to study 2-phase (air-water) flow through porous media and to demonstrate a novel "passive" method of gas-liquid separation (GLS) by means of capillary pressure gradients. We further propose the development of numerical simulation capability using Lattice-Boltzmann Modeling (LBM) methods. Numerous systems which provide thermal control and life support functions aboard ISS employ ion-exchange, adsorbent, and catalytic media in packed beds (i.e., granular or spherical porous media) through which fluids flow. Many advanced life support, thermal control, and in-situ resource utilization technologies under development for future manned lunar and Mars missions also employ porous media (e.g., Sabatier reactors, Bosch reactors, reverse water-gas-shift reactors, ilmenite reduction reactors, VPCAR, biological water processors, etc.).		

Task Description:	An improved understanding is needed of the effects of microgravity and hypogravity on multiphase flow as it relates to: 1) the physical environment (temperature, pressure, gravity, g-jitter, etc.); 2) the physical and chemical properties of the fluids (density, viscosity, interfacial surface tension, compressibility, polarity, miscibility, vapor pressure, etc.); and, 3) the chemical and physical properties of the solid media (wettability (i.e., contact angle), particle morphology, particle size distribution, particle packing, porosity, tortuosity, etc.). Advanced knowledge derived from the proposed ISS experiments, embodied in highly accurate simulation software based upon LBM numerical methods, can provide both: 1) a means to facilitate optimal design of future systems, and, 2) extensive problem solving capability, which will be particularly useful when malfunctions or other off-nominal events occur. Microgravity GLS is currently performed via membrane or centrifugal separation systems. Membranes suffer from deterioration of performance over time due chemical and microbial fouling, and the accumulation of particulate matter which interferes with trans-membrane mass transfer. Centrifugal systems require rotating machinery, which consume energy, require maintenance, and produce g-jitter. Preliminary experiments indicate the strong potential for effective "passive" GLS via capillary pressure gradients, including demonstration of sufficient force to overcome buoyancy in the 1g environment. Demonstration of this methodology will serve as an excellent ISS experiment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Advances in the understanding of multiphase flow characteristics under variable gravity conditions will ultimately lead to improved and as of yet unknown process designs for advanced space missions. Such novel processes will be of paramount importance to the success of future manned space exploration as we venture into our solar system and beyond. In addition, because of the ubiquitous nature and vital importance of biological and environmental processes involving air/water mixtures, knowledge gained about fundamental interactions and the governing properties of these mixtures will clearly benefit the quality of life here on our home planet. The techniques addressed in the current research involving multiphase transport in porous media and gas-liquid phase separation using capillary pressure gradients are also a logical candidate for a future ISS flight experiment. Importantly, the novel and potentially very accurate Lattice-Boltzmann modeling of multiphase transport in porous media under development in this program offers significantly improved predictions of real world fluid physics phenomena, thereby promoting advanced process designs for both space and terrestrial applications.
Task Progress:	This 3-year research effort has culminated in the design and testing of a zero-g demonstration prototype. Both the hydrophilic (glass) and hydrophobic (Teflon) media Capillary Pressure Gradient (CPG) cartridges prepared during the second year’s work were evaluated. Results obtained from ground testing at 1-g were compared to those obtained at reduced gravities spanning Martian (1/3-g), Lunar (1/6-g), and zero-g. These comparisons clearly demonstrate the relative strength of the CPG phenomena and the efficacy of its application to meet NASA’s unique gas-liquid separation (GLS) requirements in non-terrestrial environments. LB modeling software, developed concurrently with the zero-g test effort, was shown to accurately reproduce observed CPG driven gas-liquid separation phenomena. The design and fabrication of a micropost plate-lamina Hele-Shaw (HS) cell was performed which served as a computationally attainable geometric structure facilitating direct comparison between physical phenomena observed in our laboratory and the LB software predictions.
Bibliography Type:	Description: (Last Updated: 06/27/2012)
Papers from Meeting Proceedings	Parker JM, Jovanovic GN, Wheeler RR, Duval WM. "Lattice-Boltzmann Model Development for Capillary Pressure Gradient Gas/Liquid Phase Separation in Microgravity." Presented at the 41st International Conference on Environmental Systems (ICES 2011), Portland, Oregon, July 17-21, 2011. 41st International Conference on Environmental Systems (ICES 2011), Portland, Oregon, July 17-21, 2011. AIAA paper AIAA-2011-5196. , Jul-2011